

BFS Traversal

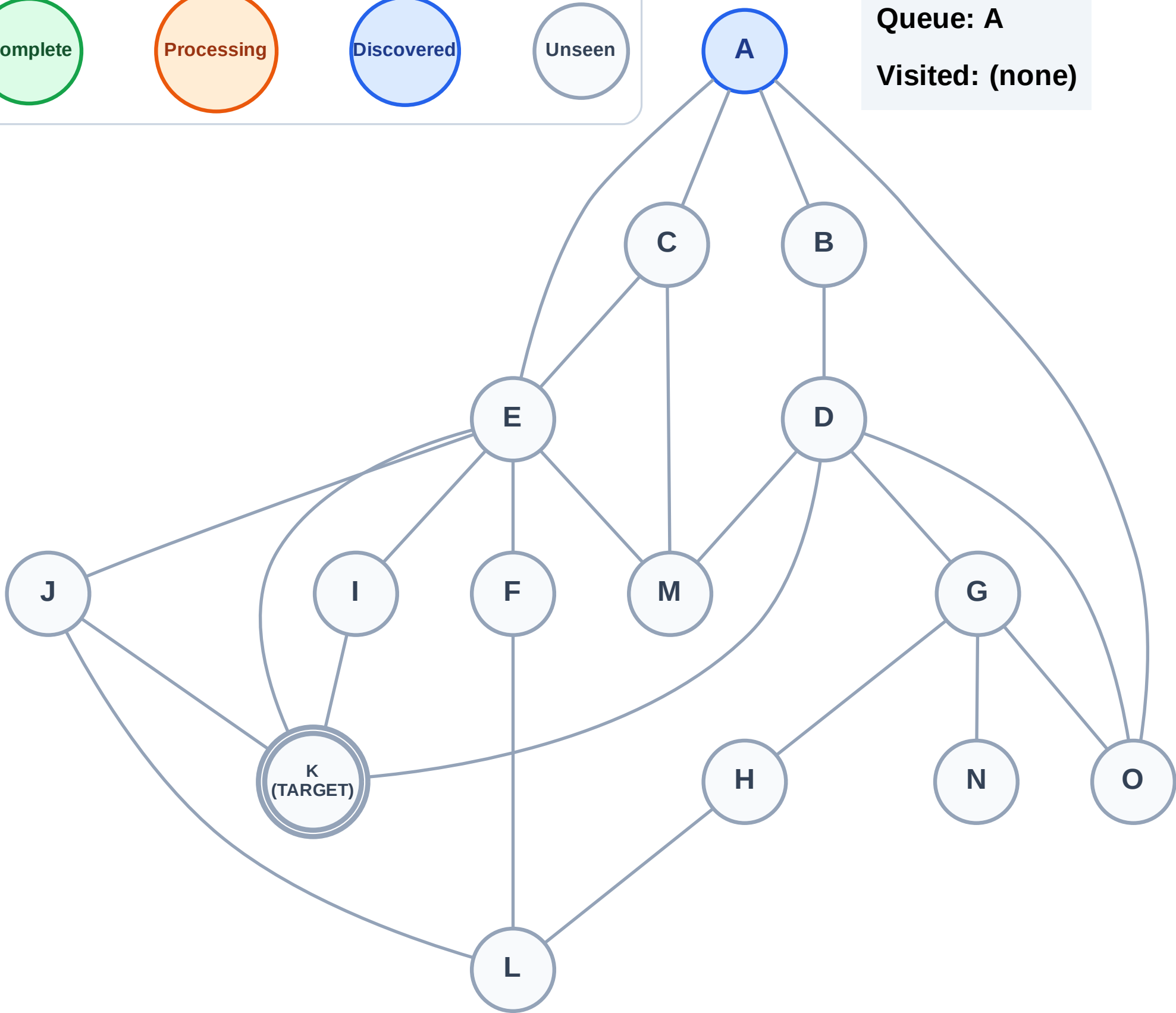
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

Visited: (none)



BFS Traversal

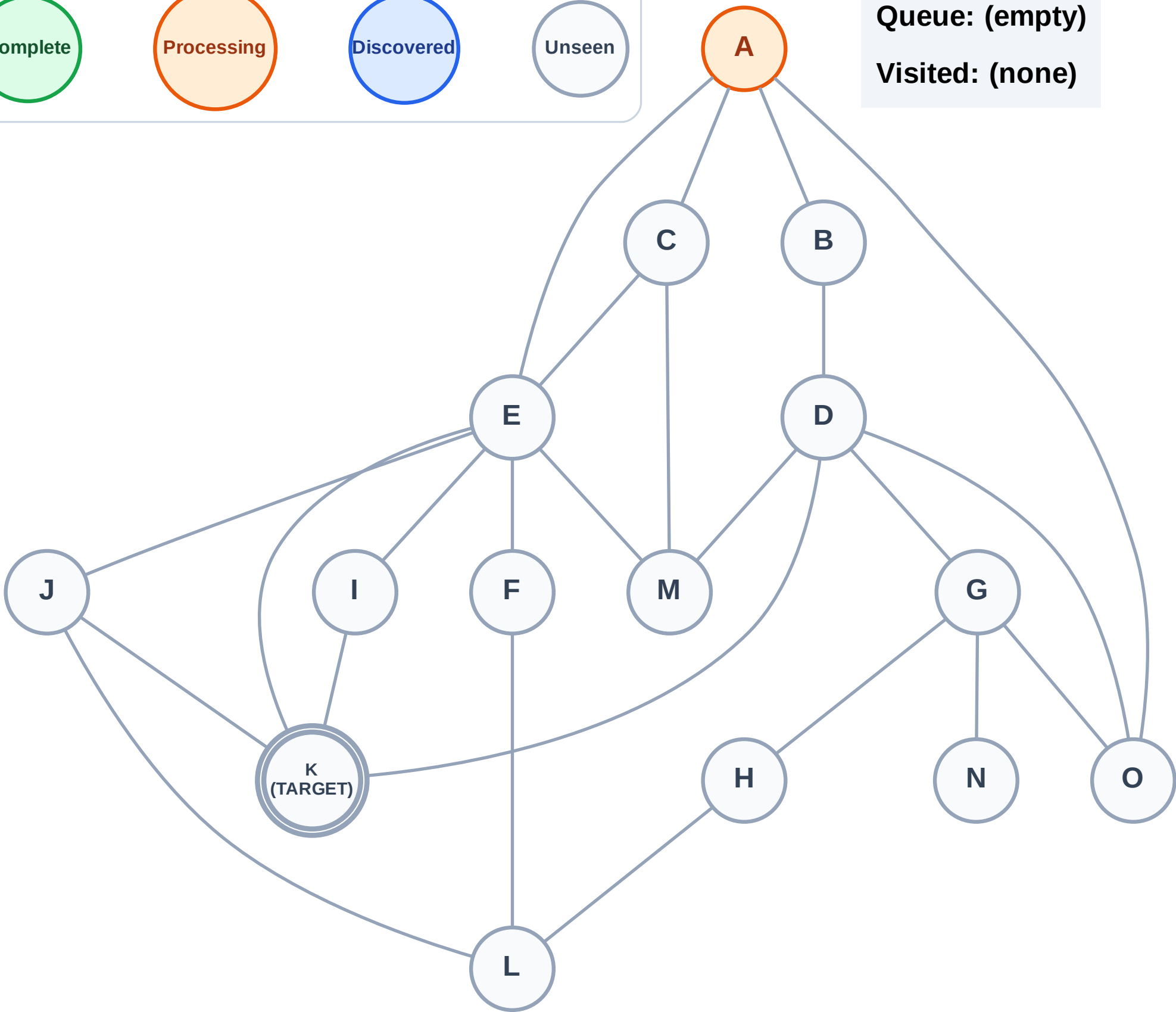
Step 2: Process node A

Legend



Queue: (empty)

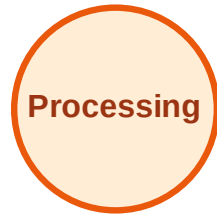
Visited: (none)



BFS Traversal

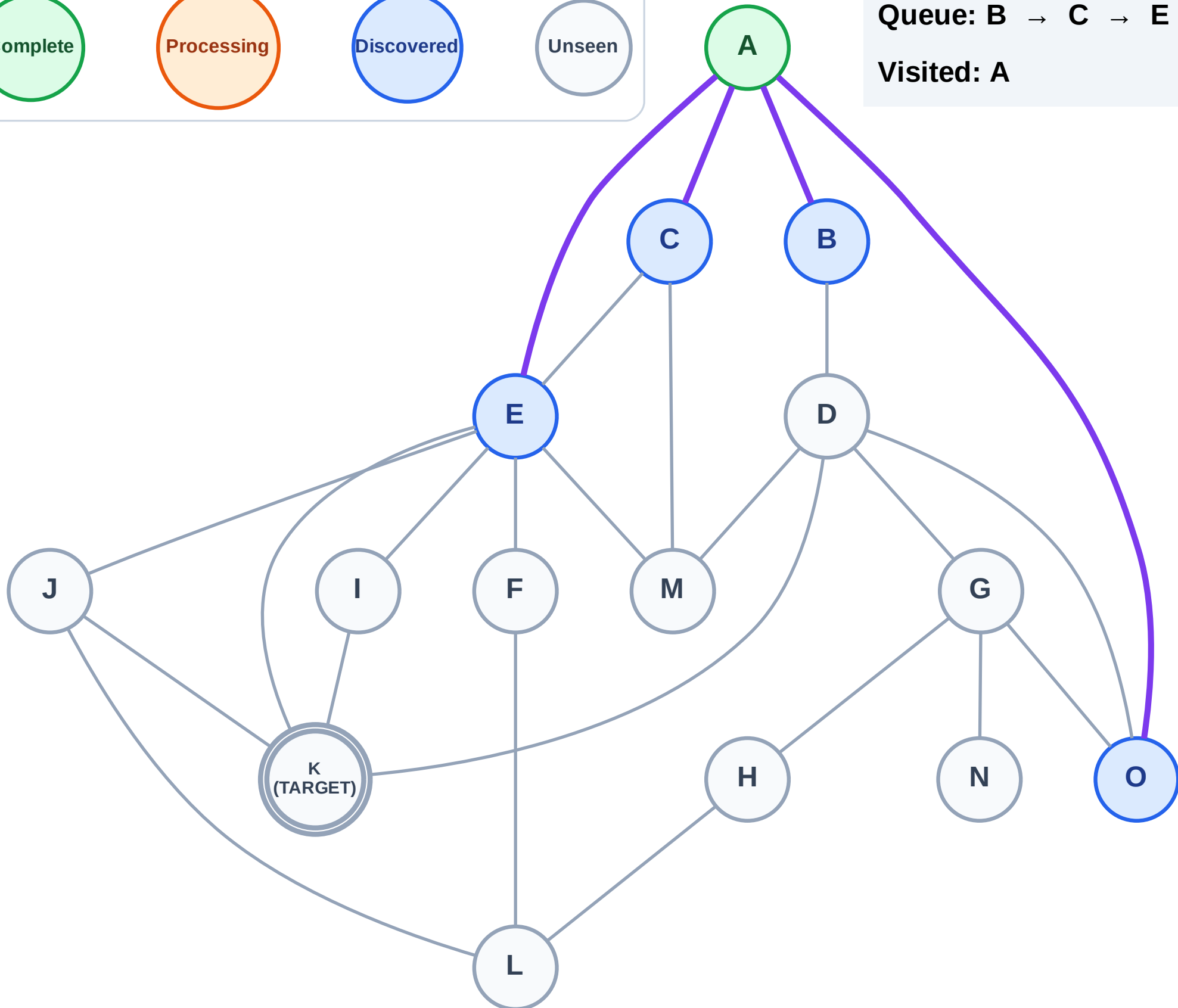
Step 3: Discover B, C, E, O from A

Legend



Queue: B → C → E → O

Visited: A



BFS Traversal

Step 4: Process node B

Queue: C → E → O
Visited: A

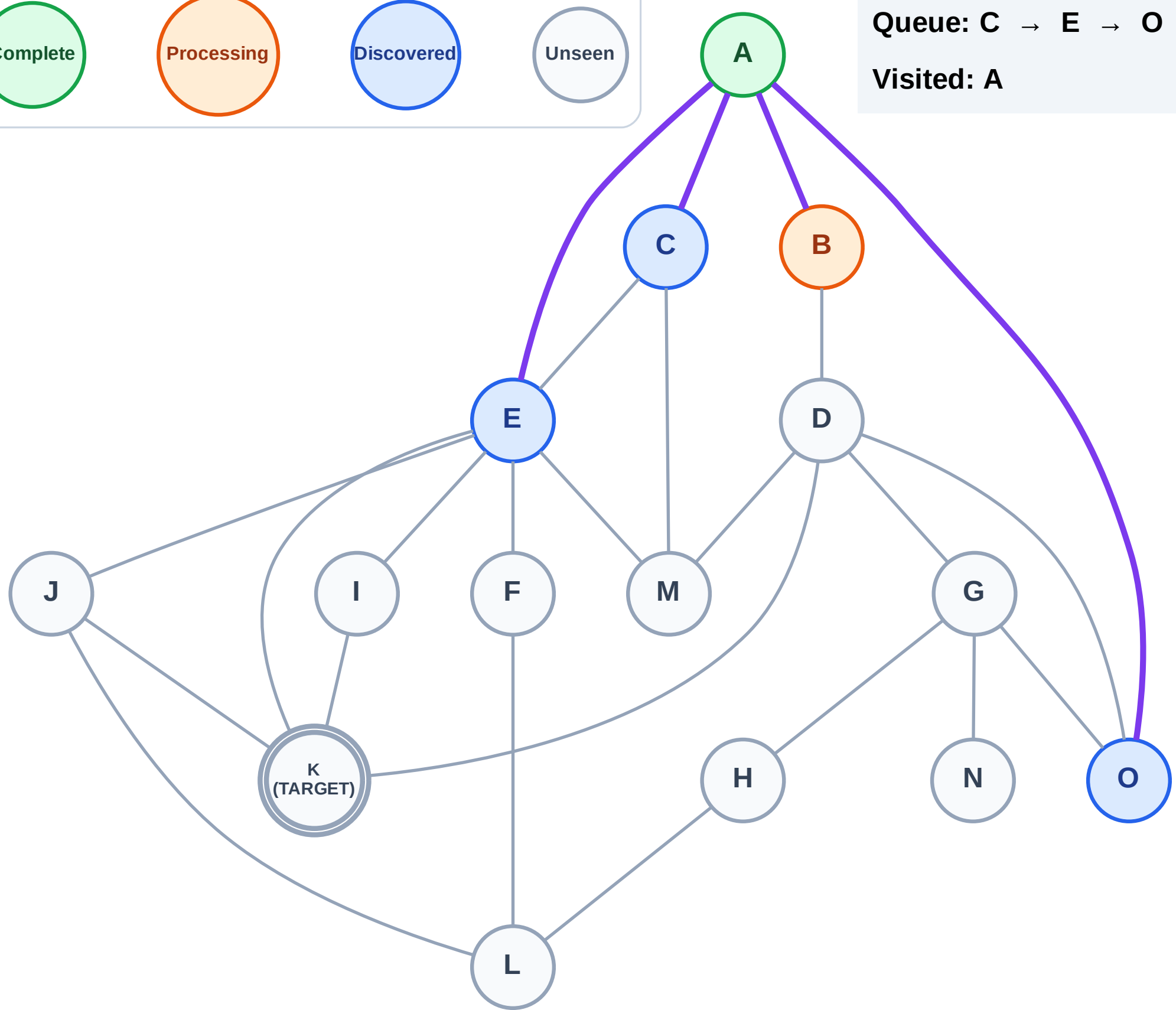
Legend

Complete

Processing

Discovered

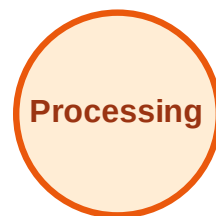
Unseen



BFS Traversal

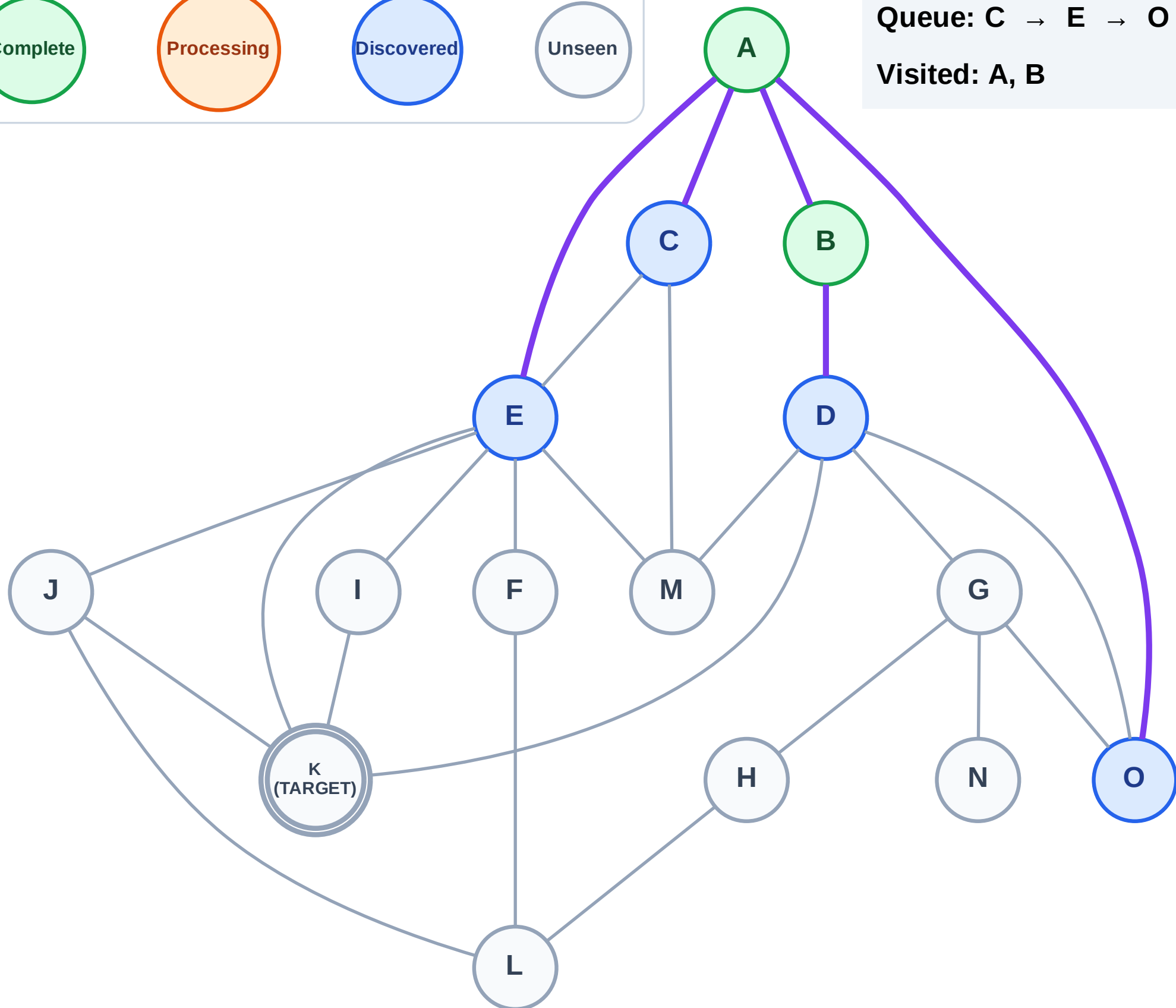
Step 5: Discover D from B

Legend



Queue: C → E → O → D

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

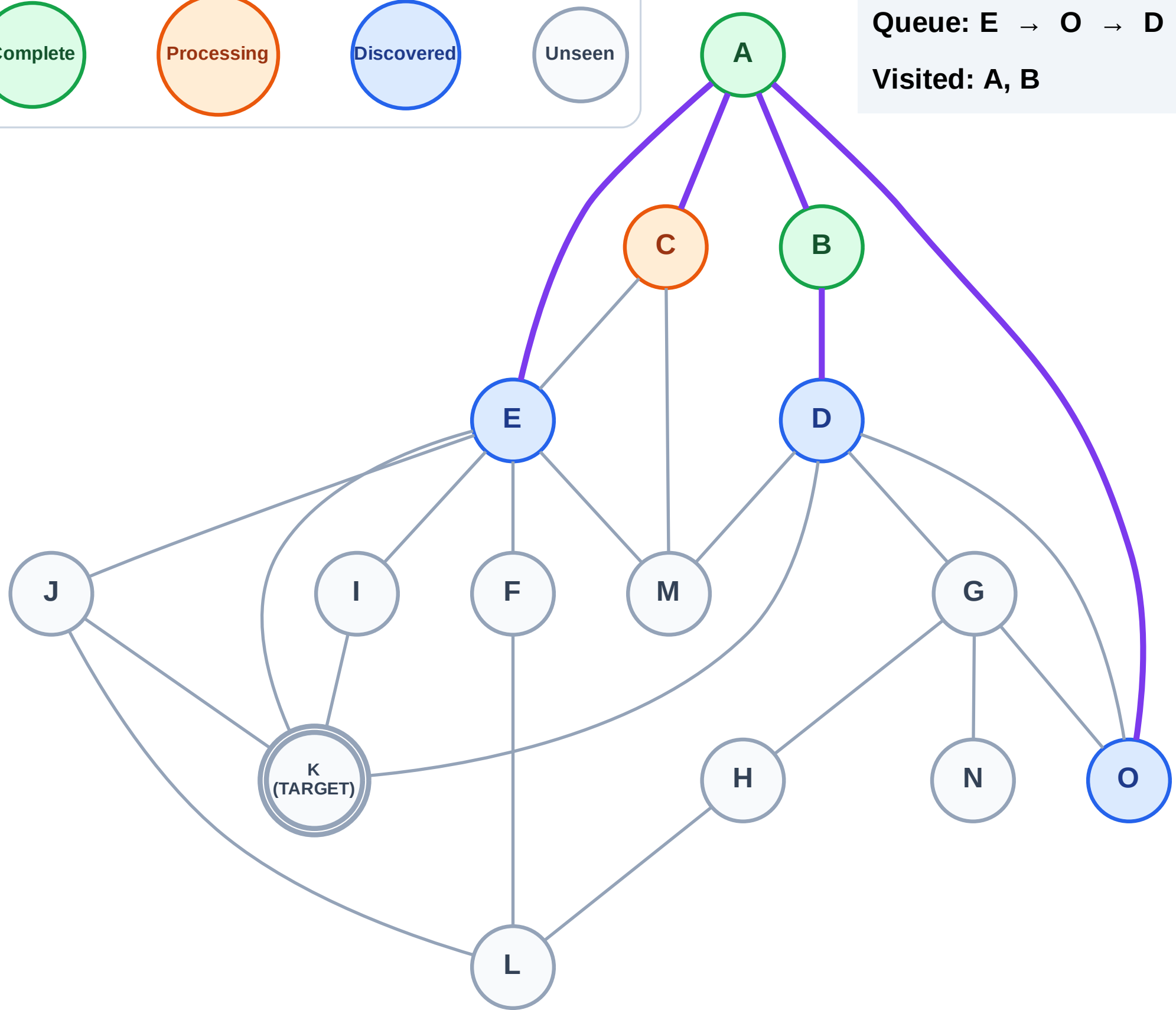
Processing

Discovered

Unseen

Queue: E → O → D

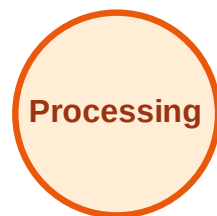
Visited: A, B



BFS Traversal

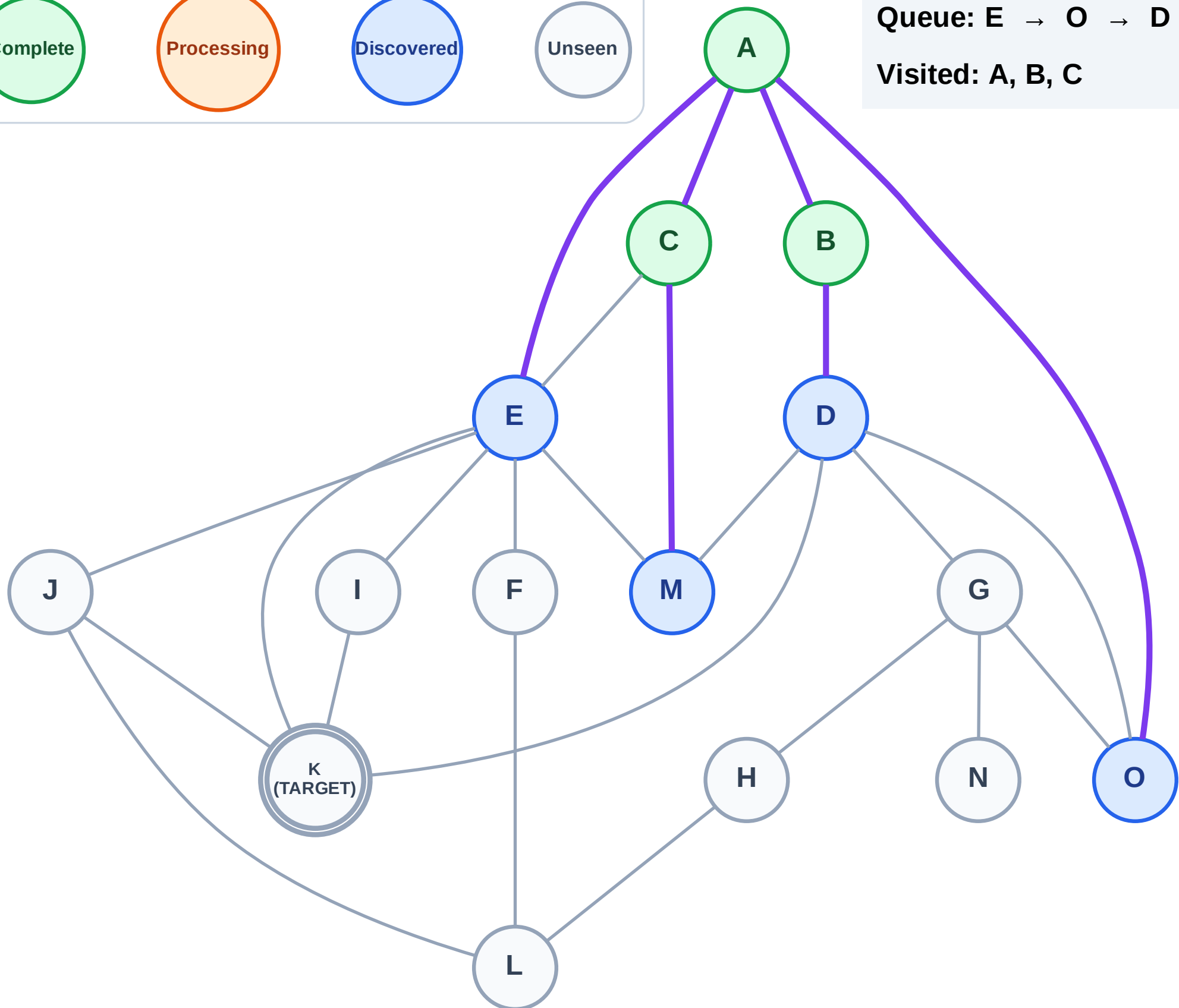
Step 7: Discover M from C

Legend



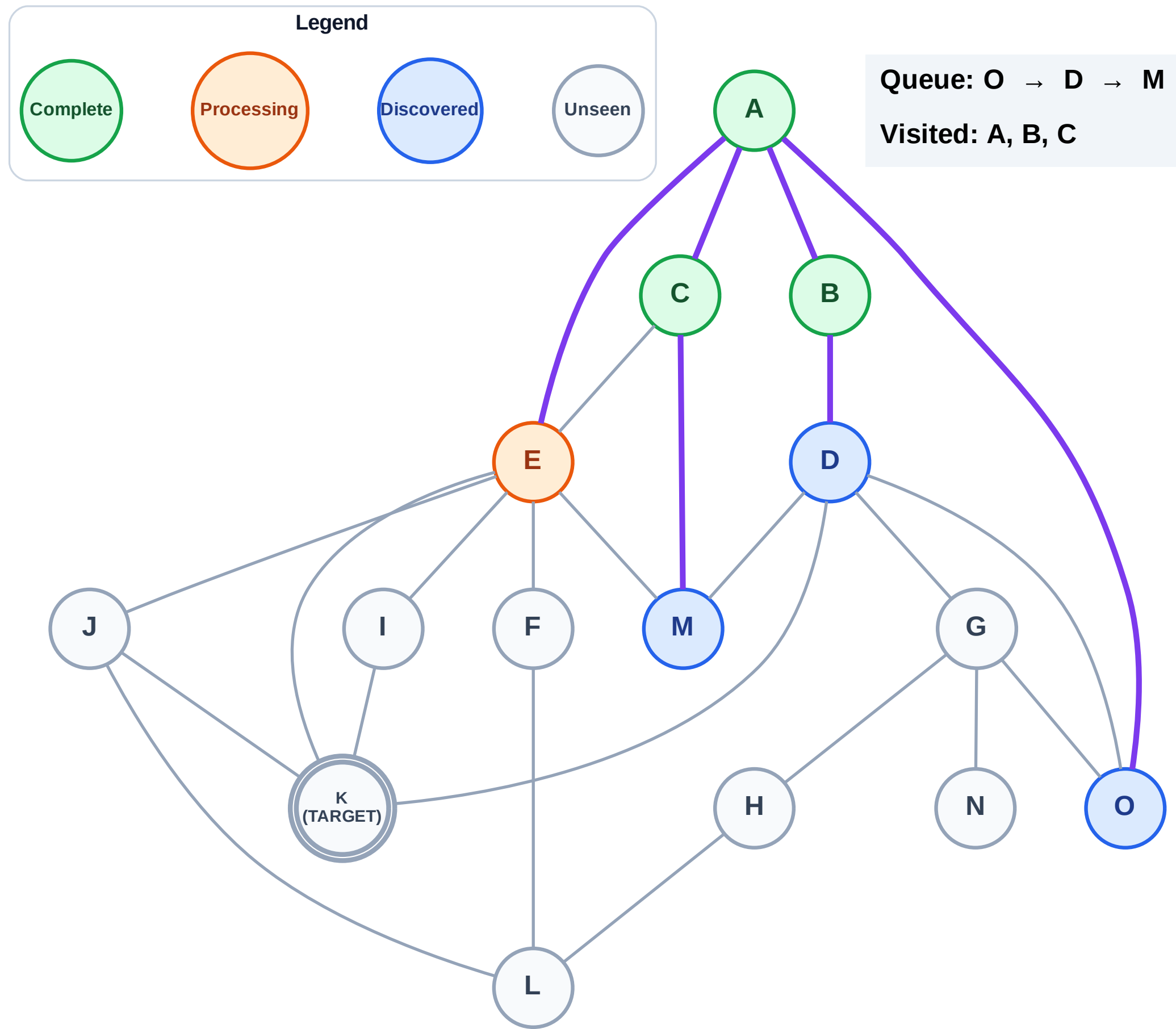
Queue: E → O → D → M

Visited: A, B, C



BFS Traversal

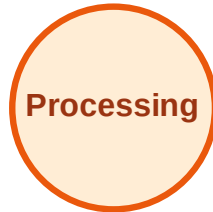
Step 8: Process node E



BFS Traversal

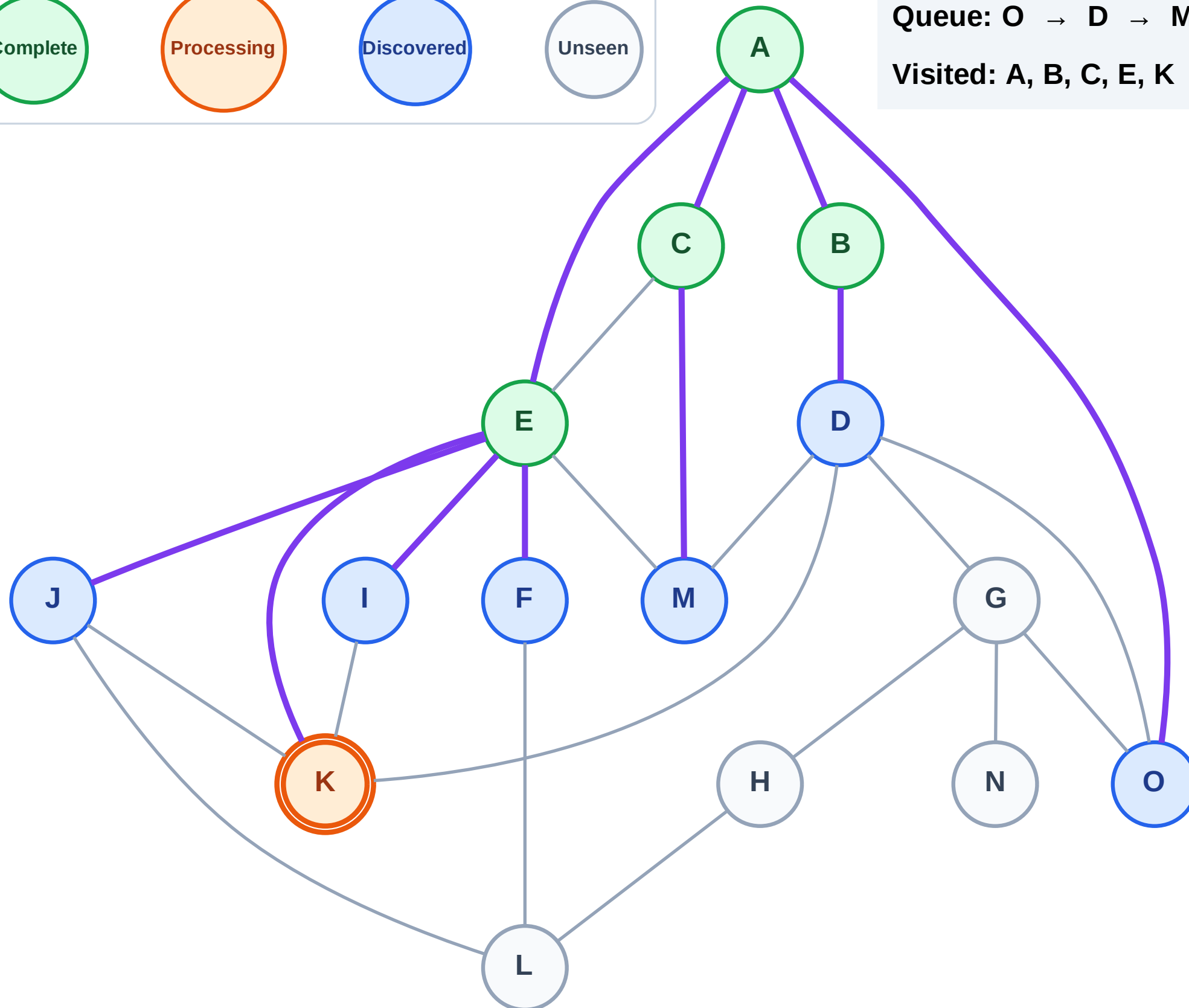
Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → E → K

Legend



Queue: O → D → M → F → I → J

Visited: A, B, C, E, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

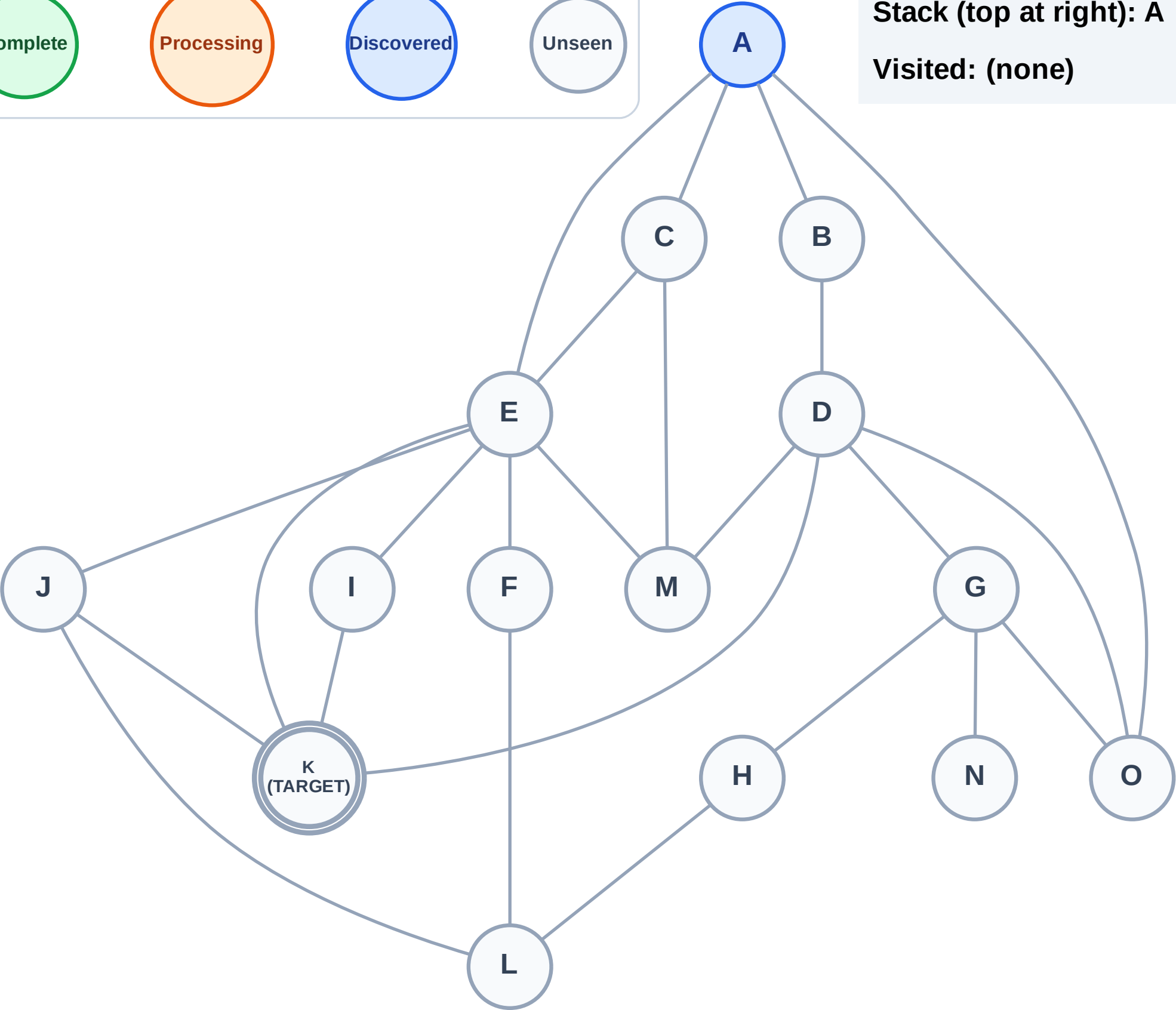
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

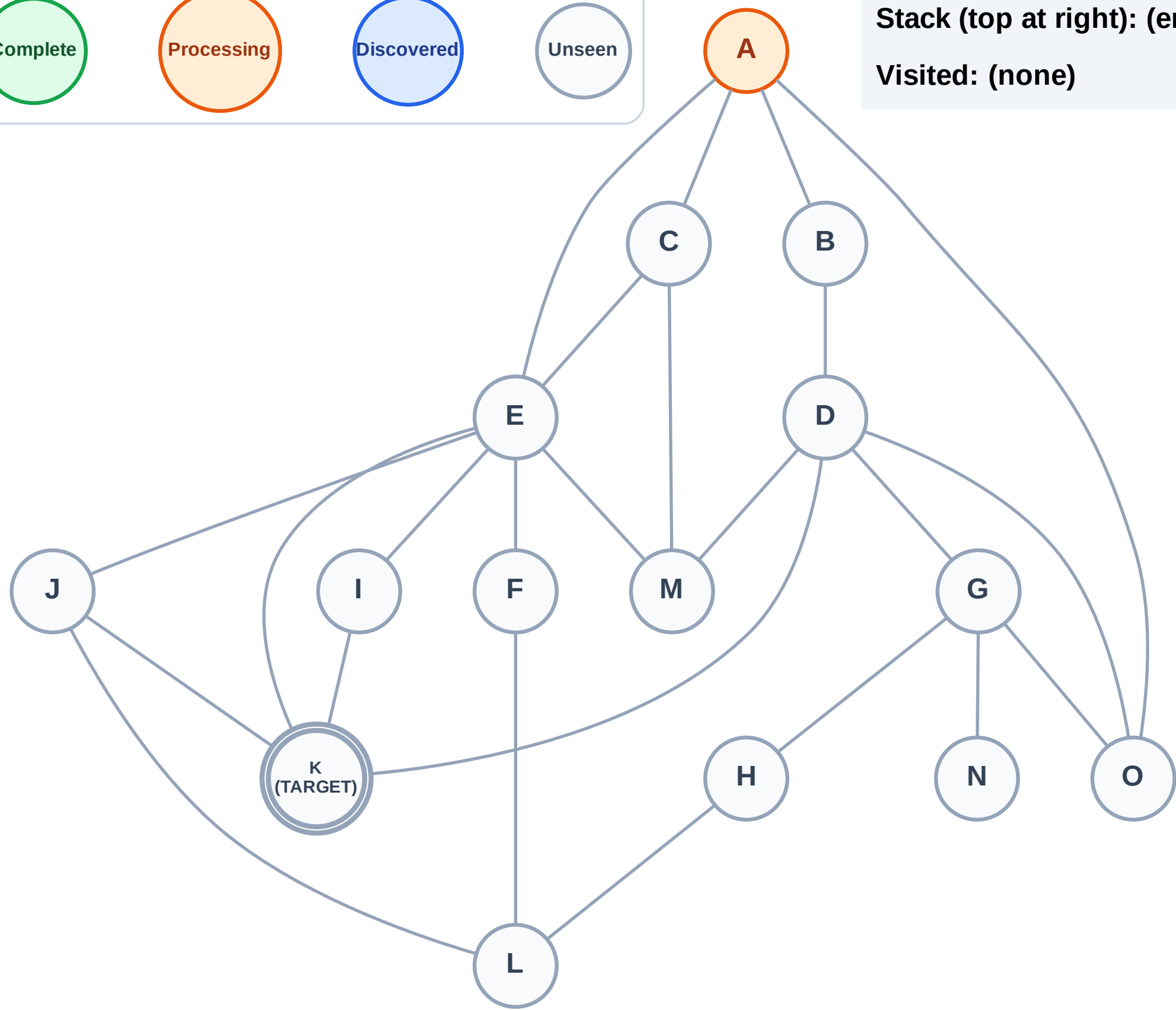
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover O, E, C, B from A

Legend

Complete

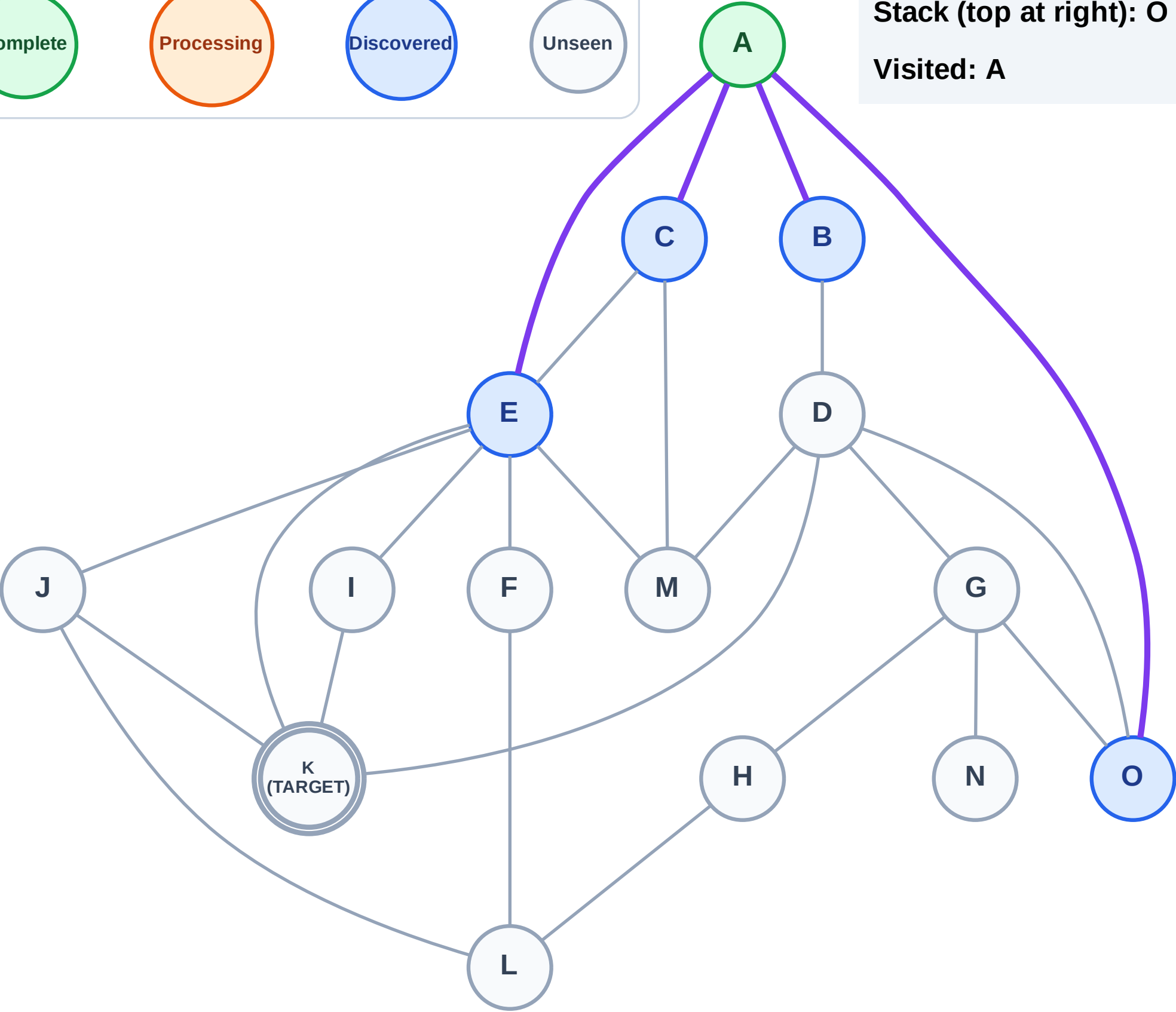
Processing

Discovered

Unseen

Stack (top at right): O | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

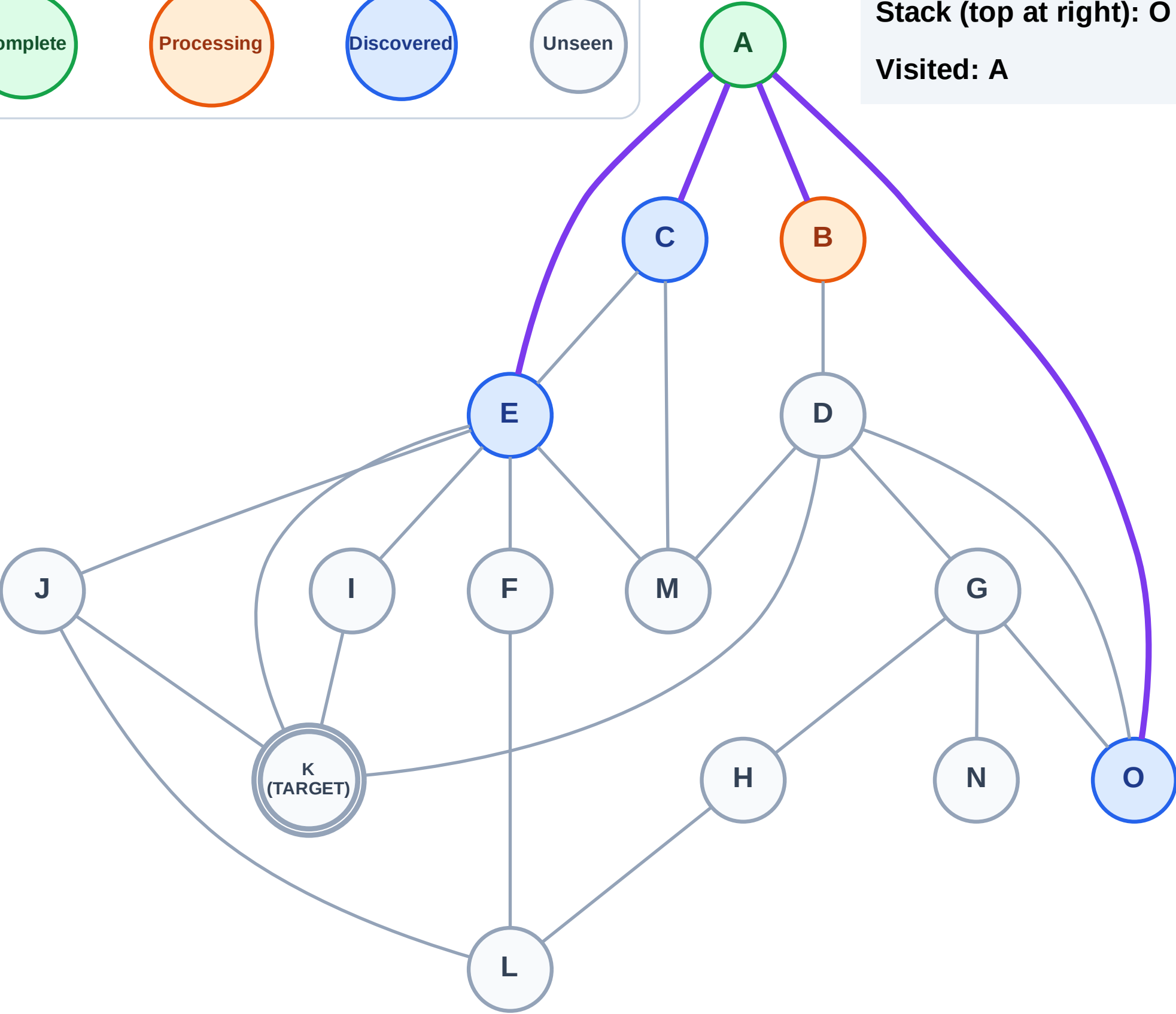
Processing

Discovered

Unseen

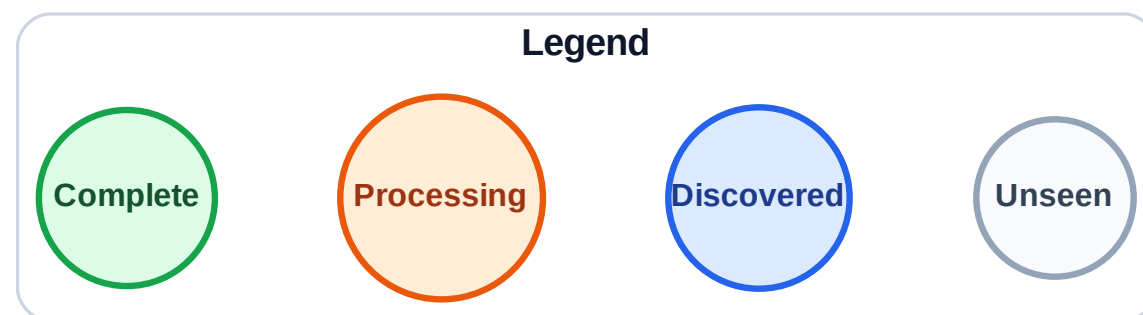
Stack (top at right): O | E | C

Visited: A

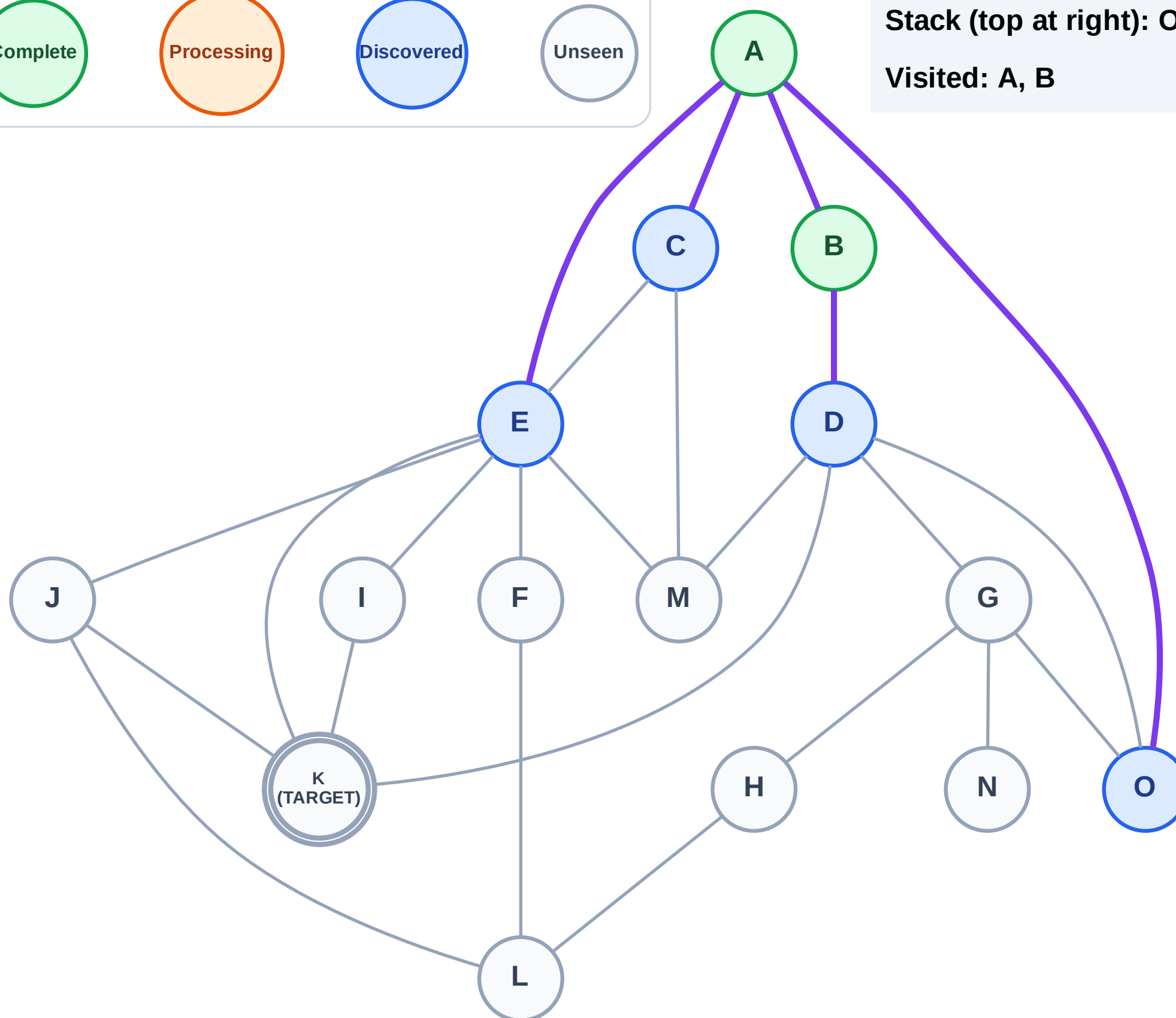


DFS Traversal

Step 5: Discover D from B



Stack (top at right): O | E | C | D
Visited: A, B



DFS Traversal

Step 6: Process node D

Legend

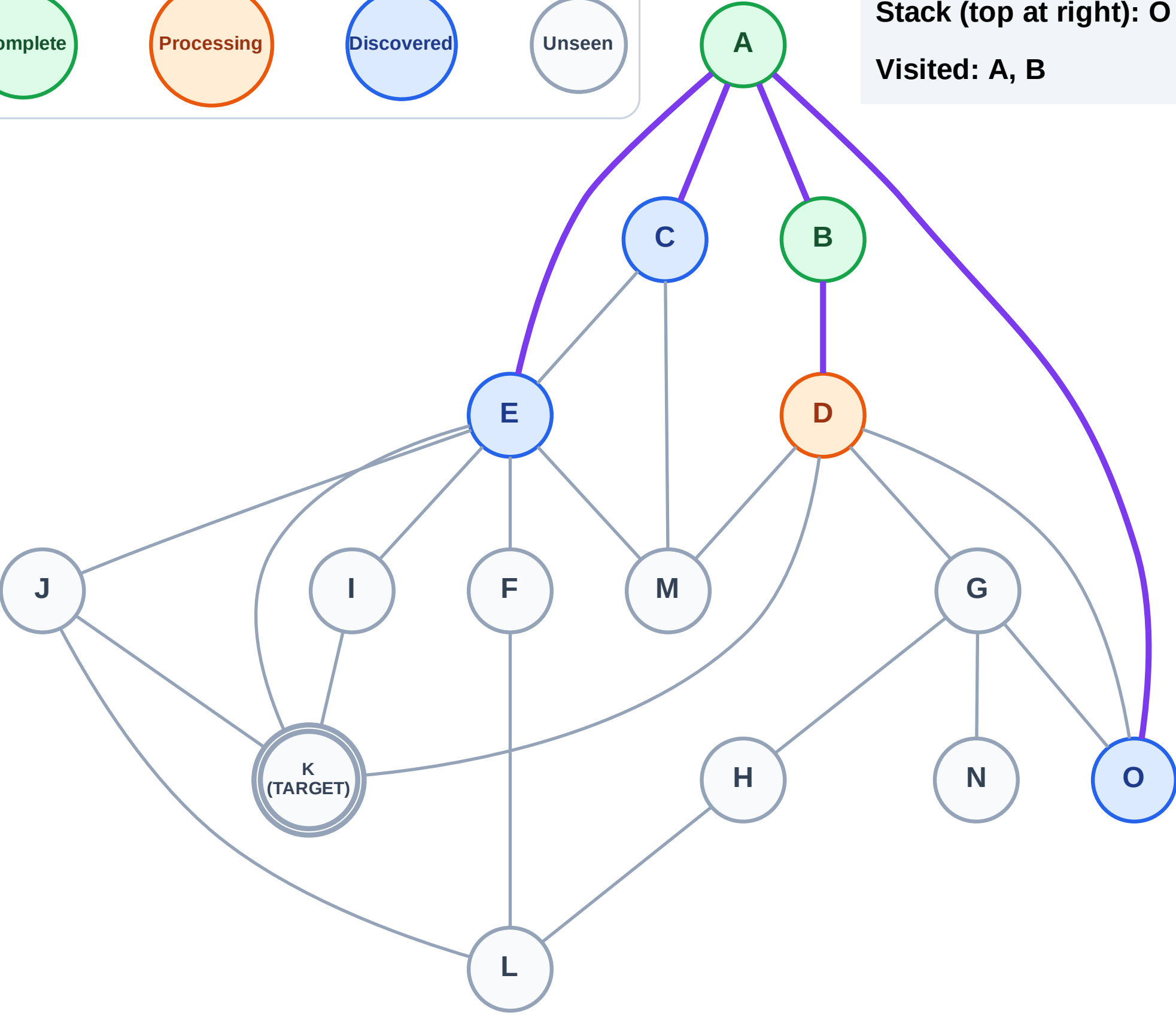
Complete

Processing

Discovered

Unseen

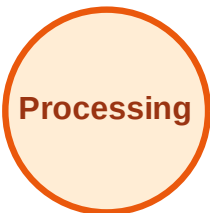
Stack (top at right): O | E | C
Visited: A, B



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → D → K

Legend



Stack (top at right): O | E | C | M

Visited: A, B, D, K

